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Top Skills

LangGraph
Multi-agent Systems
Retrieval-Augmented Generation (RAG)

Languages

English
Hindi

Certifications

IBNC CISCO Networking
Networking structure of B.C.C.L.
Python 3 Tutorial Course
Introduction to Data Science
Linux Shell Scripting, System Programming & Networking

Honors-Awards

Best Intern
Siemens You Answered
Employee Of Quater

Publications

Heart Disease Prediction System
Using Random Forest

Patents

Method and System for Generating
a Geometric Component using
Machine Learning

Yeshvendra Kumar Singh

Senior Software Developer @ Trend Micro | Ex-NOKIA | Ex-Siemens
| University of Ottawa | IIT Dhanbad | Sikkim Manipal | Passionate AI
Enthusiast

Ottawa, Ontario, Canada

Summary

Several years of experience in C\C++, NodeJS, JavaScript, Python, Qt, Mendix, etc. with hands-on industry exposure in designing, developing, and deploying scalable applications. Special interest in Machine Learning, Data Science with experience in data manipulation, processing, visualization, and usage of various machine learning algorithms. In past have worked as a Software Engineer at Siemens Digital Industries Software. Completed M.Tech, in Computer Science & Engineering from IIT (ISM) Dhanbad and now pursuing MDTI in Applied Data Science from the University of Ottawa.

Experience

Trend Micro
Senior Software Developer
July 2024 - Present (2 years 1 month)
Ontario, Canada

At Trend Micro, I work on cutting-edge cybersecurity solutions that combine cloud security, AI, and digital twin innovation to help organizations strengthen their security posture.

Attack Path Modeling Across Multi-Cloud – Creating advanced logic to detect and model potential attack paths across Azure, GCP, OCI, AWS, as well as in Data Security Posture Management (DSPM) and AI Security Posture Management (AI-SPM) domains.

AI-Driven Security Analysis – Using artificial intelligence to dynamically uncover and prioritize attack vectors, enabling proactive risk mitigation before they impact production environments.

Digital Twin for Cybersecurity – Driving the inception of digital twin concepts to replicate enterprise infrastructures, enabling organizations to simulate, validate, and predict security risks without costly real-world changes.

LLM-Powered Security Innovation – Exploring how Large Language Models (LLMs) can transform digital twins into interactive, natural language-driven platforms. My work spans across:

1. Retrieval-Augmented Generation (RAG) for knowledge-grounded threat insights
2. Multi-Agent and Agent-to-Agent systems for distributed risk analysis and automated decision-making
3. Multi-Modal architectures to integrate diverse data sources (logs, configs, diagrams, telemetry) for holistic attack path discovery

Through this work, I help organizations visualize risks, test hypotheses, and optimize security posture at scale—driving down costs while staying ahead of evolving threats.

Nokia

Software Engineer Co-op

May 2023 - April 2024 (1 year)

Kanata, Ontario, Canada

At Nokia, I had the privilege of contributing to cutting-edge projects that pushed the boundaries of technology and innovation:

Prototyped a RAG-based Language Model (LLM) utilizing OpenAI and Mistral frameworks. This revolutionary system leveraged RFCs and internal documents to respond intelligently to queries, enhancing information retrieval processes.

Spearheaded the development of a predictive test case selection model. This innovative solution applied Machine Learning techniques to analyze Code Coverage data and historical code change data, enabling the system to forecast the likelihood of test case failures with precision.

Led the initiative to transition TCL test automation code to Python, streamlining our testing processes and enhancing efficiency across projects.

These experiences not only honed my technical skills but also deepened my understanding of leveraging emerging technologies to drive impactful results. I'm proud to have been part of such groundbreaking projects at Nokia, paving the way for future advancements in the field.

Siemens Digital Industries Software

5 years 4 months

Software Engineer - Senior Technical Lead

January 2022 - September 2022 (9 months)

Pune, Maharashtra, India

Worked on the production level implementation of Design Query and Automation Service which would be used as a back end for NX Connector in Mendix.

Software Engineer - Technical Lead

September 2019 - January 2022 (2 years 5 months)

Pune Area, India

Investigated the start-up performance of NX in native mode and suggested some changes to improve it. Created POC to facilitate interaction between NX and Mendix so that Mendix web application developers can use NX Connector to solve the NX-specific use-cases by building their own web application in Mendix. Even figured out an AWS-based deployment strategy for the NX Connector Service.

Helped in support of mixed unit assemblies in NX.

Created POC for Selection Prediction project which is now in production release of NX.

Associate Software Engineer

June 2018 - August 2019 (1 year 3 months)

Pune Area, India

Created a framework for easy migration of NX users from one version of NX to another. Provided NX users with the report of what has changed in-between versions of NX. Provided Qt-based desktop tool to identify the configurations which might impact when upgrading from one version to another.

Intern

June 2017 - May 2018 (1 year)

Pune Area, India

Combined the UF-Wrapper parsers i.e. C#, Java, and Python wrapper parser into a single parser. This reduced the total number of files from 119 files to 30 files. Built a web component-based tool that would help NX developers easily analyze Syslog generated by NX.

Indian Institute of Technology (Banaras Hindu University), Varanasi Research Intern

January 2016 - May 2016 (5 months)

Varanasi, Uttar Pradesh, India

Worked on learning the basics of machine learning with hands-on experience on projects like "Prediction of Heart disease using Machine Learning", "Crack detection in ball bearing of motors using CNN", "PCA and Euclidean distance based facial recognition system", etc.

Education

University of Ottawa

Master of Digital Transformation and Innovation, Applied Data

Science · (September 2022 - April 2024)

Indian Institute of Technology (Indian School of Mines), Dhanbad

Master of Technology (M.Tech.), Computer Science and

Engineering · (2016 - 2018)

Sikkim Manipal Institute of Technology

B. Tech., C.S.E. · (2012 - 2016)

Sunbeam School Bhagwanpur

Intermediate, Computer Science, Maths, Physics, Chemistry,

English · (2010 - 2012)

St. John's D.L.W.

Highschool, science · (1998 - 2010)